

Steam Game Ownership Analysis

1. Project Overview

The project analyzes game ownership using ownership and review data from over 100,000 games from a wide variety of genres. The goal of this project is to analyze how different genres, tags, and offered languages affect the amount of owners a game has on steam, in order to determine the popularity and feasibility of certain combinations.

2. Dataset Summary

- Rows: 125,855
- Columns: 39
- Key Features:
 - Estimated Owners
 - Game Details (Tags, Languages, Genres)

3. Exploratory Data Analysis and Cleaning with Python

We began our analysis and cleaning in python:

- Data Loading: we imported the data using pandas
- Preliminary Column Check: Used `df.head()` to examine column names and watch for mismatch. There was a mismatch with the `discountDLCcount` column which we split into 2.
- Standardized Columns: we made all columns snake case for better readability and documentation
- Data exploration: we used [`df.info\(\)`](#) to check for structure and `df.describe()` for summary statistics

	appid	name	release_date	estimated_owners	peak_ccu	required_age	price	discount	dlc_count	about_the_game	...	average_playtime_two_weeks	median_playtime_forever	median_playtime_two_weeks	developers	publishers
count	1.258550e+05	125854	125855	125855	1.258550e+05	125855.000000	125855.000000	125855.000000	125855.000000	117393	...	125855.000000	1.258550e+05	125855.000000	117406	116933
unique	NaN	124644	5141	14	NaN	NaN	NaN	NaN	NaN	116776	...	NaN	NaN	NaN	72859	64607
top	NaN	Shadow of the Tomb Raider: Definitive Edition	Oct 23, 2025	0 - 20000	NaN	NaN	NaN	NaN	NaN	Help the beautiful girls to immerse themselves...	...	NaN	NaN	NaN	EroticGamesClub	BFG Entertainment
freq	NaN	20	190	75705	NaN	NaN	NaN	NaN	NaN	65	...	NaN	NaN	NaN	232	554
mean	2.032627e+06	NaN	NaN	NaN	5.318720e+01	0.165365	4.811174	17.896826	0.535434	NaN	...	13.507711	1.691803e+02	14.419125	NaN	NaN
std	1.117434e+06	NaN	NaN	NaN	3.681083e+03	1.642666	12.489329	26.628518	14.328417	NaN	...	266.925759	1.105725e+04	290.747134	NaN	NaN
min	1.000000e+01	NaN	NaN	NaN	0.000000e+00	0.000000	0.000000	0.000000	0.000000	NaN	...	0.000000	0.000000e+00	0.000000	NaN	NaN
25%	1.083495e+06	NaN	NaN	NaN	0.000000e+00	0.000000	0.590000	0.000000	0.000000	NaN	...	0.000000	0.000000e+00	0.000000	NaN	NaN
50%	1.949010e+06	NaN	NaN	NaN	0.000000e+00	0.000000	2.390000	0.000000	0.000000	NaN	...	0.000000	0.000000e+00	0.000000	NaN	NaN
75%	2.938410e+06	NaN	NaN	NaN	0.000000e+00	0.000000	5.545000	37.000000	0.000000	NaN	...	0.000000	0.000000e+00	0.000000	NaN	NaN
max	4.437540e+06	NaN	NaN	NaN	1.013934e+06	21.000000	999.980000	100.000000	3703.000000	NaN	...	20088.000000	3.429544e+06	20088.000000	NaN	NaN

categories	genres	tags	screenshots	movies
116889	117432	83353	119828	0.0
14058	2922	77184	119718	NaN

Single-player,Family Sharing	Casual,Indie	Adventure,Casual,Hidden Object	https://shared.akamai.steamstatic.com/store_it...	NaN
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23878	6912	273	20	NaN
NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN

- Data cleaning: I then used df.isnull() to check for anomalies in the data. We see that there is a name that has null value and also movies seem to only have null values. Thus, we remove the movies column and also the game entry that didn't have a name.

```

appid      0
name       1
release_date 0
estimated_owners 0
peak_ccu   0
required_age 0
price      0
discount   0
dlc_count  0
about_the_game 8462
supported_languages 0
full_audio_languages 0
reviews    113674
header_image 81
website    75335
support_url 70560
support_email 22630
windows    0
mac        0
linux      0
metacritic_score 0
metacritic_url 121597
user_score 0
positive   0
negative   0
score_rank 125815
achievements 0
recommendations 0
notes      102545
average_playtime_forever 0
average_playtime_two_weeks 0
median_playtime_forever 0
median_playtime_two_weeks 0
developers 8449
publishers 8922
categories 8966
genres     8423
tags       42502
screenshots 6027
movies     125855

```

- **Estimated Owners:** We check the games that have 0 estimated owners, most of these games seem to be demos/playtests and unpopular games. However, it is nearly impossible to look through all these games and only keep the fully released games. Thus, I chose to remove all 0 owners games. We also rewrote the owner categories to make them easier to read and understand.

estimated_owners		estimated_owners	
0 - 20000	75705	0 - 20K	75705
0 - 0	24578	20K - 50K	11397
20000 - 50000	11397	50K - 100K	5355
50000 - 100000	5355	100K - 200K	3454
100000 - 200000	3454	200K - 500K	2856
200000 - 500000	2856	500K - 1M	1155
500000 - 1000000	1155	1M - 2M	729
1000000 - 2000000	729	2M - 5M	405
2000000 - 5000000	405	5M - 10M	125
5000000 - 10000000	125	10M - 20M	51
10000000 - 20000000	51	20M - 50M	31
20000000 - 50000000	31	50M - 100M	9
50000000 - 100000000	9	100M - 200M	4
100000000 - 200000000	4		
Name: count, dtype: int64		Name: count, dtype: int64	

- **Sql Prep:** We then split the tags, languages, and genre columns into their own tables to make it easier for sql analysis

4. Data Analysis with SQL

I performed analysis using postgresql to answer a wide range of questions below are some examples of key questions:

- 1) **Total Owner Breakdown:** Total percentage of games that are in each estimated owner category

	estimated_owners text	percentage double precision
1	0 - 20K	74.75117500691181
2	20K - 50K	11.253406532643469
3	50K - 100K	5.287531103124135
4	100K - 200K	3.410482246534223
5	200K - 500K	2.8200165883328725
6	500K - 1M	1.1404478849875588
7	1M - 2M	0.7198151585765631
8	2M - 5M	0.3998973103203128
9	5M - 10M	0.12342509577787432
10	10M - 20M	0.050357439077372716
11	20M - 50M	0.030609423752912833
12	50M - 100M	0.008886606896006953
13	100M - 200M	0.003949603064891978

- 2) High Ownership Game Genres: Genres with highest percentage of games > 1 million estimated owners.

	genre_name text	total_games bigint	games_1m_plus bigint	percentage_1m_plus double precision
1	Massively Multipla...	1916	125	6.524008350730689
2	Free To Play	5860	207	3.5324232081911258
3	Action	41427	866	2.0904241195355686
4	Web Publishing	97	2	2.0618556701030926
5	Photo Editing	156	3	1.9230769230769231
6	RPG	18624	352	1.8900343642611683
7	Game Development	267	4	1.4981273408239701
8	Strategy	19847	297	1.4964478258678895
9	Animation & Mode...	410	6	1.4634146341463417
10	Video Production	290	4	1.3793103448275863

- 3) High Ownership Game Price Range: The Price range of games with highest percentage of games with >1 million estimated owners.

	price_range text	total_games bigint	games_1m_plus bigint	percentage_1m_plus double precision
1	\$20-\$40	1693	81	4.784406379208505
2	Free	9954	275	2.7627084589109905
3	\$40+	613	16	2.6101141924959217
4	\$10-\$20	9021	189	2.095111406717659
5	\$5-\$10	18303	269	1.4697044200404306
6	\$0.01-\$5	61692	520	0.8428969720547235

- 4) Highly Reviewed Games and High Ownership: What percentage of games with 90% positive reviews have >1 million estimated owners

	total_games bigint	games_1m_plus bigint	percentage_1m_plus double precision
1	28459	569	1.9993675111564004

- 5) High Ownership by Release Year: What year had the highest percentage of games released that reach > 1 million years.

	release_year numeric	total_games bigint	games_1m_plus bigint	percentage_1m_plus double precision
18	2014	1515	89	5.874587458745874
19	2015	2503	103	4.115061925689173
20	2016	4126	117	2.8356761997091615
21	2017	5903	111	1.8803997967135353
22	2018	7434	94	1.264460586494485
23	2019	7226	84	1.1624688624411845
24	2020	8725	116	1.329512893982808
25	2021	9613	98	1.0194528243004266
26	2022	10096	83	0.8221077654516641
27	2023	11671	65	0.5569359952017822
28	2024	15548	53	0.3408798559300231
29	2025	14680	32	0.21798365122615804

- 6) High Ownership Game Languages: What percentage of games in each language have reached estimated ownership of > 1 million.

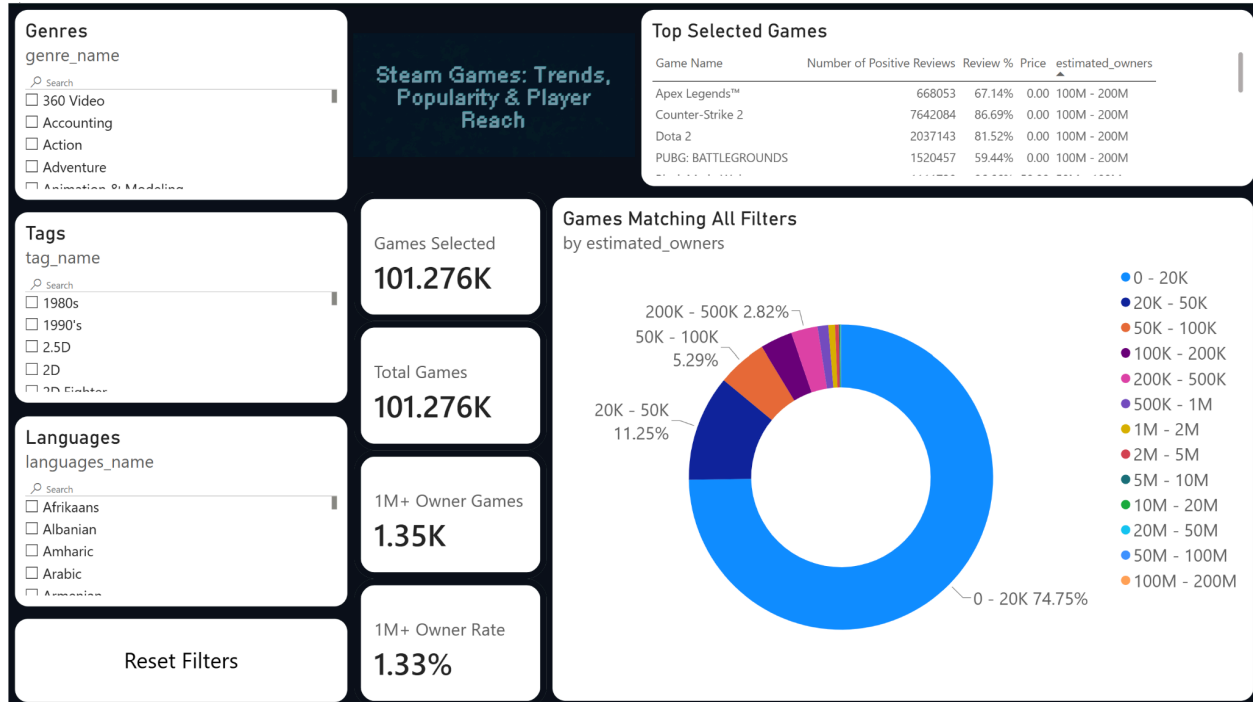
	languages_name text	total_games bigint	games_1m_plus bigint	percentage_1m_plus double precision
1	Polish	11200	716	6.392857142857143
2	Czech	4889	275	5.624872161996318
3	Italian	17847	944	5.289404381688799
4	Portuguese - Brazil	15172	726	4.785130503559188
5	Spanish - Spain	24355	1047	4.298911927735578
6	French	26128	1112	4.255970606246173
7	German	26428	1107	4.188739215983048
8	Korean	16429	676	4.1146752693407995
9	Traditional Chinese	15009	607	4.044240122593111
10	Russian	24259	936	3.8583618450884205
11	Turkish	8787	335	3.812450210538295

7) High Ownership Game Tags: Tags with the most games > 1 million estimated owners.

	tag_name text	games_1m_plus bigint
1	Singleplayer	1099
2	Action	995
3	Multiplayer	876
4	Adventure	813
5	Atmospheric	591
6	Co-op	568
7	Open World	509
8	Indie	470
9	First-Person	449
10	RPG	443

5. Dashboard in Power BI

Finally, I built an interactive Power BI dashboard to present these insights and allow us to see estimated ownership percentages for select tags, genres, and languages.



6. Recommendations

- Popular Tags: Singleplayer, Action, Multiplayer, Adventure
- Popular Languages: Polish, Czech, Italian
- Popular Genres: Massive Multiplayer, Free to Play, Action, RPG

If we were to make a singleplayer game with a small team, I would make it an action rpg adventure game to capture the biggest “market”. For languages, I would include the most used languages in the world like English, Spanish, etc. as well as Polish, Czech and Italian to include those players. I would probably price it around \$20.

If we were to make a multiplayer game with a larger team and larger budget, it would make sense to make it also action, adventure, rpg, but make it free to play and also include a wider variety of languages.